

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			A disease which is always present at low levels (in an area)/ frequently at a predictable rate in a specific location (1)	1			1		
7	(b)	(i)		A – disulphide bridge/disulphide bond B – Antigen binding site C – Light chain D – Heavy chain All 4 correct = 3 marks, 3 correct = 2 marks, 2 correct = 1 mark	3			3		
		(ii)		Antigen –initiates an {immune response/production of antibodies} (1) Antigen-antibody complex- a <u>specific</u> antibody bound to an antigen/ complementary antibody (1)	2			2		
	(c)	(i)		<u>Memory cells</u> (must make IgG during the secondary response/ after second exposure/clonal expansion)(1)		1		1		
		(ii)		Any 2 x (1) from: - The bacteria may have antigenic variation/surface proteins / reference to mutations (1) (IgM produced again as) no memory cells to new antigen (1) - Primary response each time (1)			2	2		
	(d)	(i)		Delay in detection of bacterial antigen /reference to latent period/clonal selection (1) IgG not produced straight away / IgG non existent until day 7/ Time is needed to produce and secrete antibodies (1)		2		2		